# Participation Information

Thank you for your interest in our scientific study ("Study"). Below, we inform you about the study so that you can freely decide whether you would like to participate.

**a) Project Participants**

The research project is led by:

Florian Grabs
E-Mail: florian.grabs@s2018.tu-chemnitz.de

In addition to the project management mentioned above, the following persons are involved in conducting the study:

Janet Siegmund (Scientific Staff)
E-Mail: janet.siegmund@informatik.tu-chemnitz.de

Belinda Schantong (Scientific Staff)
E-Mail: belinda.schantong@informatik.tu-chemnitz.de

Norbert Siegmund (Scientific Staff)
E-Mail: norbert.siegmund@informatik.uni-leipzig.de

If you have any questions or difficulties, you can contact the project participants at any time, even after the study.

**b) Background and objective of the study**

Developing, understanding, or debugging software is a complex and highly individual process. In addition to applying specific programming techniques, programmers are constantly confronted with solving problems. Various personal strategies and approaches are used to do this. Psychology has researched and cataloged a variety of problem-solving strategies, from trial and error to goal-means analysis. However, there has been insufficient research to date on which of these basic psychological strategies programmers actually use in practice and to what extent these depend on the respective phase in the software development cycle. The aim of this study is to examine which specific strategies programmers use in different situations of the software development cycle and which of the underlying problem-solving strategies known in psychology can be assigned to these applied techniques.

**c) Study Procedure**

Location of the study: The study takes place online.

Procedure:

Once you have agreed to participate, you will be asked a few questions to help us assess your programming experience. You will then evaluate a series of statements that describe your approach to the programming process. Based on your answers to specific tasks in the programming process, we will attempt to assess your profile. Based on your answers, we will show you a derived profile of your problem-solving strategies (with percentages). Finally, you will evaluate the extent to which this calculated profile corresponds to your own assessment.

Duration of the Study: The total duration of the study is about
30 minutes.

**d) Your Advantages and Disadvantages; Risks**

Advantages: With your participation, you contribute to human sciences research in the field of Software Engineering. With your help, problems in programming can be better understood in the future. No further personal advantages result for you from this.

Disadvantages, Risks: The study takes up about 30 minutes of your time. No recognizable risks are associated with the measurement.

**e) Inclusion and Exclusion Criteria for Participation**
You can participate in the study if you are at least 18
years old.

**f) Funding of the Research Project**
The project is financed from the following sources
Budgetary funds: Professur Softwaretechnik, Technische Universität Chemnitz

**g) Type and Extent of Compensation**
You participate in the study without compensation.

**h) Voluntariness and Right of Withdrawal**
Your participation is voluntary. You can abort the study at any time without giving reasons and without any disadvantage to you.

**i) Data Protection**
In the context of the study with the above-mentioned title, data about you will be collected, evaluated, and stored. The project participants commit to complying with data protection regulations. The handling of your data is described in the information sheet "Data Protection Information".

# CONSENT TO STUDY PARTICIPATION

I hereby declare that I have been fully informed about the study, in particular about its goals, its procedure including duration, and about the advantages and disadvantages as well as risks that might be associated with the participation. I have read and understood the participation information. All my questions have been answered to my satisfaction. I was able to download a copy of the participation information and this declaration of consent for study participation. I had enough time to consider my decision regarding study participation and to make it freely. I am aware that I can withdraw my consent to participate in the study at any time and without giving reasons, without any disadvantage resulting for me.

**I voluntarily participate in the above-mentioned study.**

Note: The consent is given by clicking a corresponding button.